

Установка и настройка GNS3

Лабораторная работа №4

Чилеше Лупупа

13 декабря 2025

Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Установка и настройка системы моделирования компьютерных сетей GNS3 и сопутствующего программного обеспечения для подготовки лабораторного стенда.

Выполнение работы

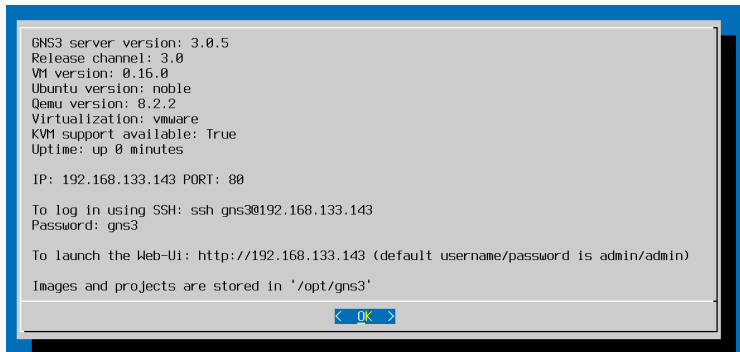
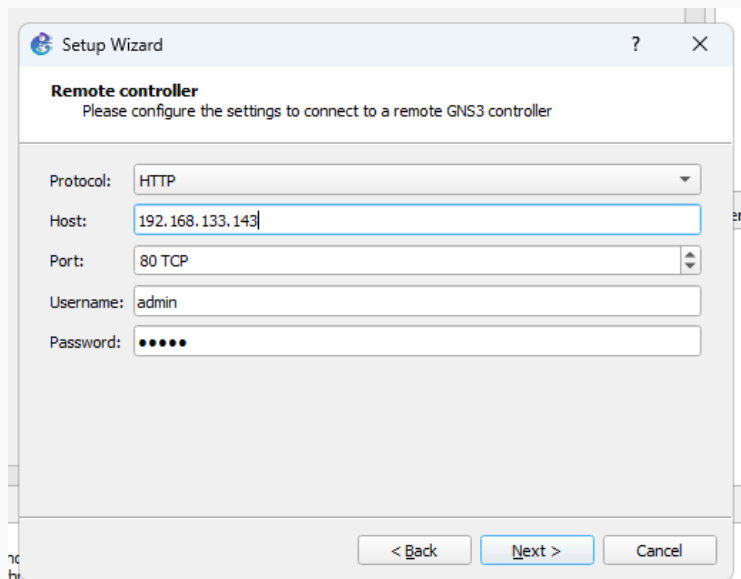


Рис. 1: Информация о состоянии GNS3 VM

Подключение к удалённому серверу GNS3



The screenshot shows a 'Setup Wizard' window with the title bar containing a question mark and a close button. The main heading is 'Remote controller' in bold, followed by the instruction 'Please configure the settings to connect to a remote GNS3 controller'. Below this, there are five input fields: 'Protocol' is a dropdown menu set to 'HTTP'; 'Host' is a text box containing '192.168.133.143'; 'Port' is a spinner box set to '80 TCP'; 'Username' is a text box containing 'admin'; and 'Password' is a text box with six dots. At the bottom right, there are three buttons: '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a blue border.

Setup Wizard

Remote controller
Please configure the settings to connect to a remote GNS3 controller

Protocol: HTTP

Host: 192.168.133.143

Port: 80 TCP

Username: admin

Password: ••••••

< Back Next > Cancel

Выбор шаблона FRR

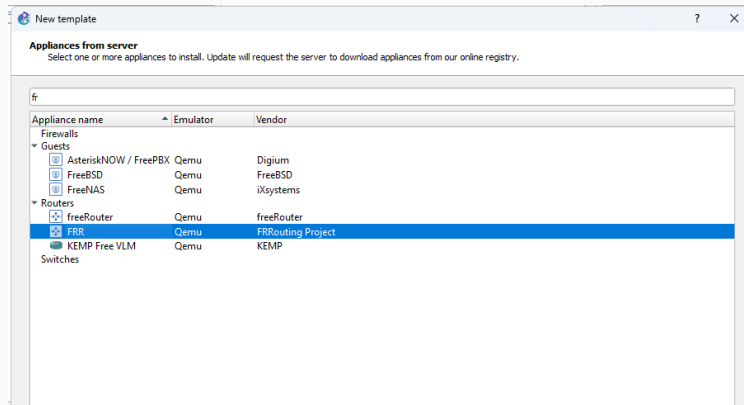


Рис. 3: Выбор шаблона FRR

Установка образа FRR

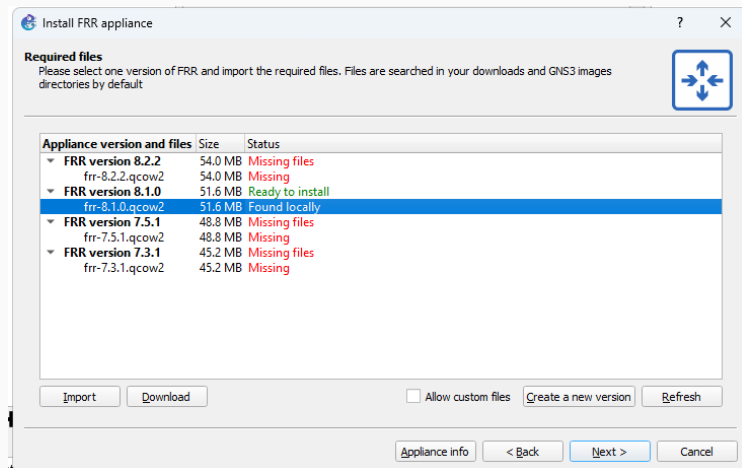


Рис. 4: Выбор версии FRR

Запуск маршрутизатора FRR

FRR-1



```
FRR-1 - PuTTY
* Setting keymap ... [ ok ]
* Starting networking ... *   lo ... [ ok ]
* Starting busybox syslog ... [ ok ]
* Initializing random number generator ... [ ok ]
* Starting busybox acpid ... [ ok ]
* Starting busybox crond ... [ ok ]
Started watchfrr
* Starting sshd ... [ ok ]

Welcome to Alpine!

The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <http://wiki.alpinelinux.org/>.

You can setup the system with the command: setup-alpine

You may change this message by editing /etc/motd.

Hello, this is FRRouting (version 8.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

=> frr#
```

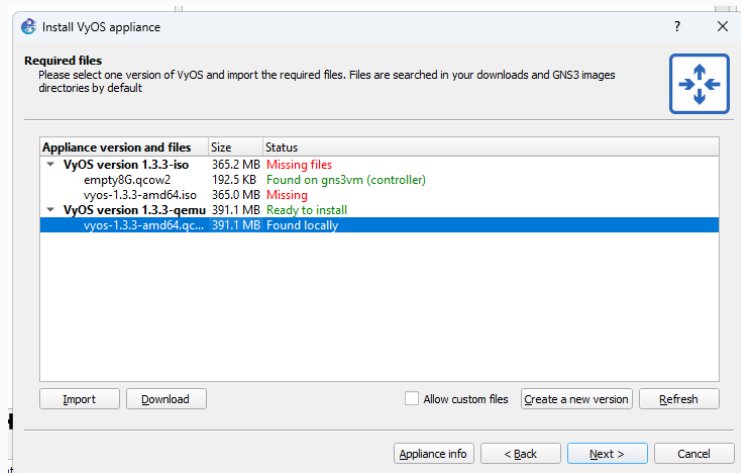


Рис. 6: Выбор версии VyOS

Запуск маршрутизатора VyOS

VyOS-1



VyOS-1 - PuTTY

```
[ 13.944458] EXT4-fs (sdal): re-mounted. Opts: (null)
[ 14.413416] random: crng init done
[ 14.576127] systemd[1]: Inserted module 'autofs4'
[ 14.594971] systemd[1]: systemd 241 running in system mode. (+PAM +AUDIT +SEL
INUX +IMA +APPARMOR +SMACK +SYSVINIT +UTMP +LIBCRYPTSETUP +GCRYPT +GNUTLS +ACL +
XZ +LZ4 +SECCOMP +BLKID +ELFUTILS +KMOD -IDN2 +IDN -PCRE2 default-hierarchy=hybr
id)
[ 14.604782] systemd[1]: Detected virtualization kvm.
[ 14.607480] systemd[1]: Detected architecture x86-64.
[ 14.620371] systemd[1]: Set hostname to <vyos>.
[ 15.073241] systemd[1]: Listening on udev Control Socket.
[ 15.077308] systemd[1]: Reached target Swap.
[ 15.081196] systemd[1]: Listening on Journal Socket (/dev/log).
[ 15.085425] systemd[1]: Listening on initctl Compatibility Named Pipe.
[ 15.089822] systemd[1]: Reached target Network.
[ 15.094271] systemd[1]: Created slice system-serial\x2dgetty.slice.
[ 15.204281] bridge: filtering via arp/ip/ip6tables is no longer available by
default. Update your scripts to load br_netfilter if you need this.
[ 15.215189] Bridge firewalling registered
[ 15.234837] mpls_gso: MPLS GSO support

Welcome to VyOS - vyos ttyS0

vyos login: █
```

Вывод

- Установлена и настроена среда моделирования GNS3 с использованием GNS3 VM
- Выполнено подключение клиента к удалённому серверу
- Успешно импортированы и запущены маршрутизаторы FRR и VyOS
- Лабораторный стенд полностью готов к дальнейшей работе